

END TERM EXAMINATION

THIRD SEMESTER [B.TECH] JANUARY 2024

Paper Code: ECC-207 Subject: Digital Logic & Computer Design

Time: 3 Hours Maximum Marks: 75

Note: Attempt five questions in all including Q.No.1 which is compulsory.

Select one question from each unit. Assume missing data, if any.

Q1 Attempt all questions:

(3×5=15)

- Reduce the Boolean Expression $+^-(+^-)$
- What are don't care terms?
- Differentiate a decoder from a Demultiplexer.
- Draw the state table and excitation table of T flip flop.
- Write about Virtual memory.

UNIT-I

- Q2 a) What is meant by duality in Boolean algebra? State and prove Associative and Distributive theorems. (7)
- b) Minimize the expression $(, , ,) = \sum (0,1,2,5,8,9,10)$ using K-map and obtain (i) minimal SOP and (ii) minimal POS expression. (8)
- Q3 a) Explain the operation of a magnitude comparator? (7)
- b) Implement a full subtractor using 8:1 MUX. (8)

UNIT-II

- Q4 a) What is Race Around condition? And how is it overcome in JK Flip-Flop? Explain with diagram. (5)
- b) Design a Mod-5 synchronous counter using JK Flip-Flop. (10)
- Q5 a) Explain designing of T Flip-Flop using JK Flip-Flop. (7)
- b) Write short notes on: (i) Shift registers (ii) PLA & PAL (8)

UNIT-III

- Q6 a) Differentiate between hardwired control and micro programmed control and explain organizations of micro programmed control unit with neat diagram. (10)
- b) What is addressing mode? Explain different types of addressing modes. (5)
- Q7 a) What are the different phases of an instruction cycle? Explain with the help of flowchart. (7)
- b) What is assembly language? Explain basic computer instruction formats with suitable diagram. (8)

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UNIT-IV

- Q8 a) Define DMA. Explain its need. Explain DMA transfer in detail with the help of suitable diagram. (7)
- b) What is an interrupt? Explain interrupt cycle with the help of flowchart. (8)
- Q9 Write short notes on:
- Modes of data transfer (5)
 - Cache memory (5)
 - Associative memory (5)
